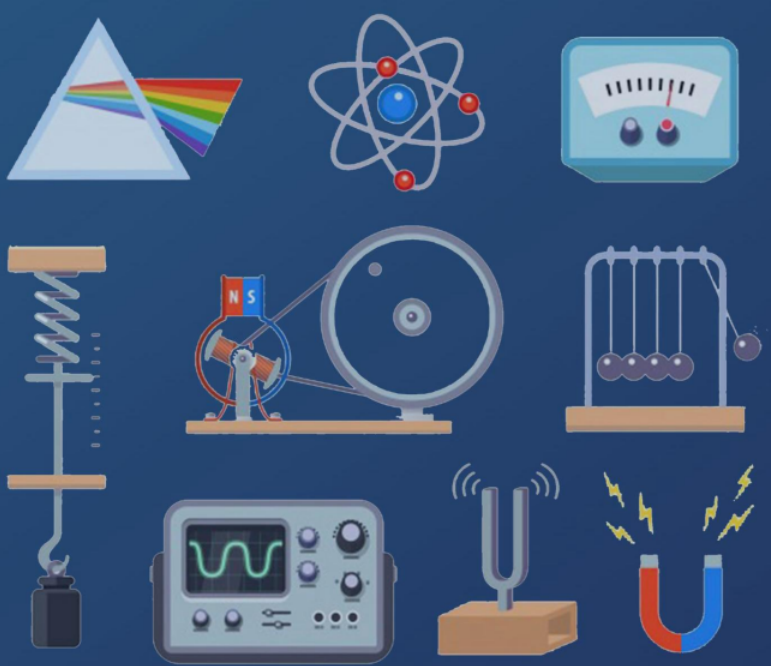
PHYSICS

Work, Power & Energy



SHEET

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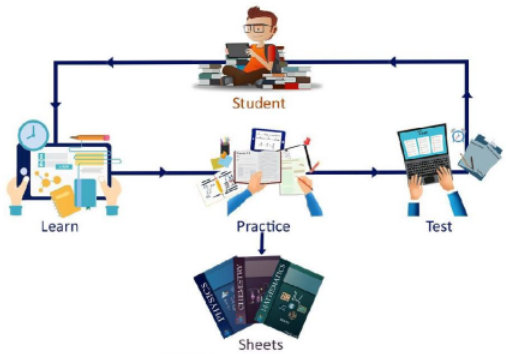
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Work - Introduction	1-5	19	1	
2	Work done by Constant and Variable Force	6,7	15,20,26	2 to 5	1,13,14, 27-30
3	Work done by Spring			6,7	31
4	Work Energy Theorem for a Particle	8-13	1,16	8-15	2,32
5	Vertical Circular Motion	14	17	16,17	3, 4, 15-19, 22
6	Work Energy Theorem for a System	15-16	21	18-25	20,21,23, 33,37,38
7	Central and Tangential Forces		2		5
8	Conservative Forces	17	3,22	26	6
9	Potential Energy	18,19	18,23	27,28	
10	Mechanical Energy and its Conservation	20-24	4	29 to 38	26
11	Equilibrium	25			7
12	Power	26-28	24	39,40	39,40
13	After Full Chapter		5-14, 25, 27		8-12, 24-25, 34-36, 41

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction

Q1. A bicyclist comes to a skidding stop in 10 m. During this process, the force on the bicycle due to the road is 200N and is directly opposed to the motion. The work done by the cycle on the road is ?

(A) + 2000J (B) – 200J

(C) zero (D) – 20,000J

Q2. A rope is used to lower vertically a block of mass M by a distance x with a constant downward acceleration g/2. The work done by the rope on the block is :

(A) Mgx (B) $\frac{1}{2}$ Mgx

(C) $-\frac{1}{2}$ Mgx (D) Mgx^2

Q3. A block of mass 5 kg slides down a rough inclined surface. The angle of inclination is 45°. The coefficient of sliding friction is 0.20. When the block slides 10 m; the work done on the block by force of friction is :-

(A) $50\sqrt{2}$ J (B) $-50\sqrt{2}$ J

(C) 50 J (D) –50 J

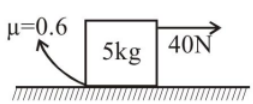
Q4. A 40N force start acting on a particle of mass 5kg, which is resting on a rough surface. If co-efficient of friction between block and surface is $\mu = 0.6$. Find work done by firction in 3 seconds :-

(A) –90 J

(B) –60 J

(C) –180 J

(D) –270 J



Q5. A body moves a distance of 10 m along a straight line under the action of a force 5 N. If the work done is 25 joules, the angle which the force makes with the direction of motion of the body is :-

(A) 0° (B) 30°

(C) 60° (D) 90°

Work done by Constant and Variable Force

Q6. A force of magnitude of 30 N acting along $\hat{i}+\hat{j}+\hat{k}$, displaces a particle from point (2, 4, 1) to (3, 5, 2). The work done during this displacement is -

(A) 90 J (B) 30 J

(C) $30\sqrt{3}$ J (D) $30/\sqrt{3}$ J.

Q7. $\vec{F}=(\hat{i}+\hat{j})$ newton, displacement $\vec{S}=(3\hat{i}+4\hat{j})$ metre work done is :-

(A) 3J (B) 4J (C) 7J (D) 5J

Work Energy Theorem for a Particle

Q8. Work done in time t on a body of mass m which is accelerated from rest to a speed v in time t_1 as a function of time t is given by:

(A) $\frac{1}{2}m\frac{v}{t_1}t^2$ (B) $m\frac{v}{t_1}t^2$

(C) $\frac{1}{2}\left(\frac{mv}{t_1}t\right)^2 t^2$ (D) $\frac{1}{2}m\frac{v^2}{t_1^2}t^2$

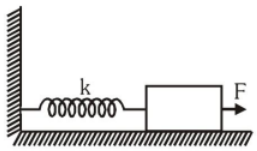
Q9. A ball with mass m projected horizontally off the end of a table with an initial kinetic energy K. At a time t after it leaves the end of the table it has kinetic energy 3K. What is t? Neglect air resistance.

(A) $(3/g)\sqrt{K/m}$ (B) $(2/g)\sqrt{K/m}$

(C) $(1/g)\sqrt{8K/m}$ (D) $(K/g)\sqrt{6/m}$

Q10. A block attached to a spring, pulled by a constant horizontal force, is kept on a smooth surface as shown in the figure. Initially, the spring is in the natural state. Then the maximum positive work that the applied force F can do is :

[Given that spring does not break]



(A) $\frac{F^2}{k}$ (B) $\frac{2F^2}{k}$ (C) ∞ (D) $\frac{F^2}{2k}$

Q11. A particle of mass 100 g is thrown vertically upwards with a speed of 5 m/s. The work done by the force of gravity during the time the particle goes up is -

(A) – 0.5 J (B) –1.25 J

(C) 1.25 J (D) 0.5 J

Q12. A constant force of 5 N is applied on a block of mass 20 kg for a distance of 2.0 m, the kinetic energy acquired by the block is :-

(A) 20 J (B) 15 J

(C) 10 J (D) 5 J

Q13. A particle of mass 2kg travel as per $x = \frac{t^3}{3}$ where x & t are in m & s. Total work done in 0 to 2 sec is :-

(A) 1.6 J (B) 16 J

(C) 160 J (D) 1600 J

Vertical Circular Motion

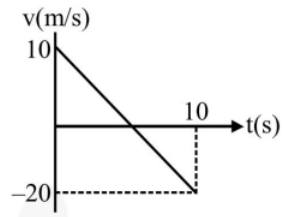
Q14. A particle of mass m is fixed to one end of a light rigid rod of length ℓ and rotated in a vertical circular path about its other end. The minimum speed of the particle at its highest point must be :

(A) zero (B) $\sqrt{g\ell}$

(C) $\sqrt{1.5g\ell}$ (D) $\sqrt{2g\ell}$

Work Energy Theorem for a System

Q15. Velocity time graph of a particle moving in a straight line is as shown in figure. Mass of the particle is 2kg. Work done by all the forces acting on the particle in time interval between t = 0 to t = 10 s is :-



(A) 300 J (B) –300 J

(C) 400 J (D) –400 J

Q16. Two bodies of masses 1 kg and 5 kg are dropped gently from the top of a tower. At a point 20 cm from the ground, both the bodies will have the same :-

(A) Momentum

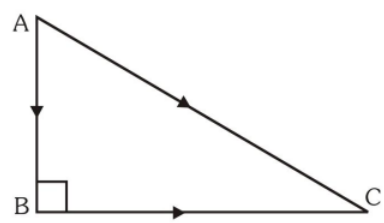
(B) Kinetic energy

(C) Velocity

(D) Total energy

Conservative Forces

Q17. The work done in moving a particle under the effect of a conservative force, from position A to B is 3 joule and from B to C is 4 joule. The work done in moving the particle from A to C is :



(A) 5 joule (B) 7 joule

(C) 1 joule (D) –1 joule

Potential Energy

Q18. The potential energy of a body is given by, $U = A - Bx^2$ (Where x is the displacement). The magnitude of force acting on the particle is :-

(A) Constant

(B) Proportional to x

(C) Proportional to x^2

(D) Inversely proportional to x

Q19. The potential energy for a force field $\vec{F}$ is given by $U(x,y) = \cos(x + y)$. The force acting on a particle at position given by coordinates $\left(0, \frac{\pi}{4}\right)$ is :-

(A) $-\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$ (B) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$

(C) $\left(\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}\hat{j}\right)$ (D) $\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}\right)$

Mechanical Energy and its Conservation

Q20. A marble of mass m and radius b is placed in a hemispherical bowl of radius r. The minimum velocity to be given to the marble so that it reaches the highest point is :

(A) $\sqrt{2g(r-b)}$ (B) $\sqrt{2gr}$

(C) $\sqrt{2g(r+b)}$ (D) $\sqrt{g(r-b)}$

Q21. A pendulum bob has been pulled aside from its equilibrium position through an angle α and then released. If ℓ is the length of the pendulum,then the bob will reach the equilibrium position with speed of :-

(A) $\sqrt{2g\ell}$ (B) $\sqrt{2g\ell\cos\alpha}$

(C) $\sqrt{2g\ell(1-\cos\alpha)}$ (D) $\sqrt{2g\ell(1-\sin\alpha)}$

Q22. When a conservative force does positive work on a body :

(A) The potential energy increases

(B) The potential energy decreases

(C) Total energy increases

(D) Total energy decreases

Q23. When a conservative force does positive work then potential energy of the system associated with that force.

(A) decreases

(B) increases

(C) remain constant

(D) depends on whether other non-conservative force is working or not.

Q24. A ball of mass 0.2 kg is thrown vertically upwards by applying a force by hand. If the hand moves 0.2 m while applying the force and the ball goes upto 2m height further, find the magnitude of force.

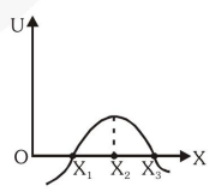
[Take $g = 10\ m/s^2$]

(A) 22N (B) 4N

(C) 16 N (D) 20 N

Equilibrium

Q25. In the figure shown the potential energy (U) of a particle is plotted against its position 'x' from origin. Then which of the following statement is correct. A particle at :



(A) x_1 is in stable equilibrium

(B) x_2 is in stable equilibrium

(C) x_3 is in stable equilibrium

(D) None of these

Power

Q26. A body of mass m accelerates uniformly from rest to v_1 in time t_1. The instantaneous power delivered to the body as a function of time t is-

(A) $\frac{mv_1t}{t_1}$ (B) $\frac{mv_1^2t}{t_1^2}$

(C) $\frac{mv_1t^2}{t_1}$ (D) $\frac{mv_1^2t}{t_1}$

Q27. An elevator with passengers has a total mass of 800 kg and moves slowly a vertical distance of 20.0 m in 10.0 s. What is the average power expended in lifting this mass?

(A) 1.57×10^2 W (B) 1.57×10^4 W

(C) 1.57×10^5 W (D) 1.57×10^6 W

Q28. A force of 10 N is applied on a body for 3 s and the corresponding displacement is 6m. The power of the source is :-

(A) 20 W (B) 25 W

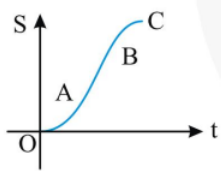
(C) 40 W (D) 50 W

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Displacement time graph of a particle moving in a straight line is as shown in figure. select the correct alternative (s):



(A) Work done by all the forces in region OA and BC is positive.

(B) Work done by all the forces in region AB is zero.

(C) Work done by the forces in region BC is negative.

(D) Work done by all the forces in region OA is negative.

Q2. Which of the following is/are conservative force(s)?

(A) $\vec{F} = 2r^3\hat{r}$ (B) $\vec{F} = -\frac{5}{r}\hat{r}$

(C) $\vec{F} = \frac{3(x\hat{i} + y\hat{j})}{(x^2+y^2)^{3/2}}$ (D) $\vec{F} = \frac{3(y\hat{i} + x\hat{j})}{(x^2+y^2)^{3/2}}$

Q3. If one of the forces acting on a particle is conservative then :

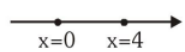
(A) Its work is zero when the particle moves exactly once around any closed path.

(B) Its work equals the change in the kinetic energy of the particle.

(C) It obeys Newton's second law.

(D) Its work depends on the end points of the motion, not on the path between.

Q4. A particle of mass m = 1 kg lying on x–axis experiences a force given by law F = x(3x–2) Newton, where x is the x–coordinate of the particle in meters. The points on x–axis where the particle is in equilibrium are :



(A) x = 0 (B) x = 1/3

(C) x = 2/3 (D) x = 1

Q5. A jumper jumps upwards with some acceleration. Choose the **CORRECT** statement (s) :-

(A) The force exerted by the ground on him while he is attempting to jump is greater than his weight.

(B) Work is done by normal force on him while he attempts to jump.

(C) Since the feet touching ground remain at rest while he is attempting to jump, force exerted by ground on him is equal to his weight.

(D) Since the feet touching ground remain at rest while he is attempting to jump, work done by the force exerted by ground on him is zero.

Q6. Choose the correct alternative(s) :

(A) When a conservative force does positive work, the potential energy of the system decreases.

(B) Work done by friction on a body always results in a loss of its kinetic energy.

(C) Work done in the motion of a body over a closed loop is zero for every force in nature.

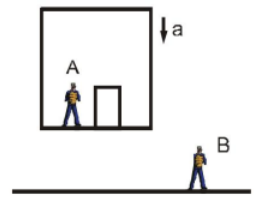
(D) Work done by friction on a body may increase its kinetic energy.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 7 to 11

A block of mass m is kept in an elevator which starts moving downward with an acceleration a as shown in figure. The block is observed by two observers A and B for a time interval t_0.



Q7. The observer B finds that the work done by gravity is

(A) $\frac{1}{2}mg^2t_0^2$ (B) $-\frac{1}{2}mg^2t_0^2$

(C) $\frac{1}{2}mgat_0^2$ (D) $-\frac{1}{2}mgat_0^2$

Q8. The observer B finds that work done by normal reaction N is

(A) zero (B) $-Nat_0^2$

(C) $+\frac{Nat_0^2}{2}$ (D) None of these

Q9. The observer B finds that work done by pseudo force is

(A) zero (B) $-ma^2t_0$

(C) $+ ma^2t_0$ (D) $-mgat_0$

Q10. According to observer B, the net work done on the block is

(A) $-\frac{1}{2}ma^2t_0^2$ (B) $\frac{1}{2}ma^2t_0^2$

(C) $\frac{1}{2}mgat_0^2$ (D) $-\frac{1}{2}mgat_0^2$

Q11. According to the observer A

(A) The work done by gravity is zero

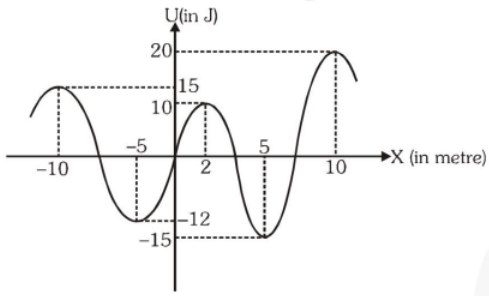
(B) the work done by normal reaction is zero

(C) The work done by pseudo force is zero

(D) All the above

Paragraph for Question no. 12 to 14

In the figure the variation of potential energy of a particle of mass m=2kg is represented w.r.t. its x-coordinate. The particle moves under the effect of this conservative force along the x- axis.



Q12. If the particle is released at the origin then :
(A) It will move towards positive x-axis.
(B) It will move towards negative x-axis.
(C) It will remain stationary at the origin.
(D) Its subsequent motion cannot be decided due to lack of information.

Q13. If the particle is released at $x = 2 + \Delta$ where $\Delta \to 0$ (it is positive) then its maximum speed in subsequent motion will be-

(A) $\sqrt{22}$ m/s (B) $\sqrt{25}$ m/s

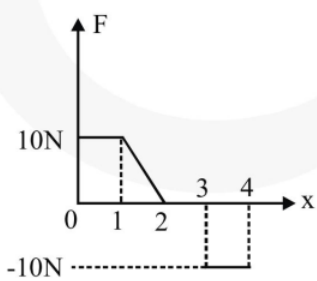
(C) $\sqrt{24}$ m/s (D) $\sqrt{23}$ m/s

Q14. x = –5 m and x = 10 m positions of the particle are respectively of-
(A) Neutral and stable equilibrium
(B) Neutral and unstable equilibrium
(C) Unstable and stable equilibrium
(D) Stable and unstable equilibrium.

MATCH THE COLUMN TYPE

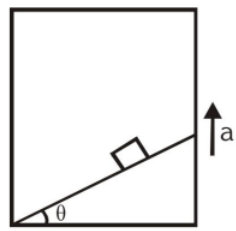
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. A variable force whose variation is shown in force–position graph below. For each of the position indicated in column-I, match appropriate description in column-II. The force is acting on a body of mass 1 kg initially at rest at x = 0.



Column I	Column II
(A) $0 < x < 1$	(P) speed is increasing
(B) $1 < x < 2$	(Q) work done by the force is positive
(C) $2 < x < 3$	(R) body moves in positive x-direction
(D) $3 < x < 4$	(S) speed is decreasing

Q16. A block of mass m is stationary with respect to a rough wedge as shown in figure. Starting from rest, in time t work done on the block : ($m = 1kg$, $\theta = 30°$, $a = 2m/s^2$, $t = 4s$)

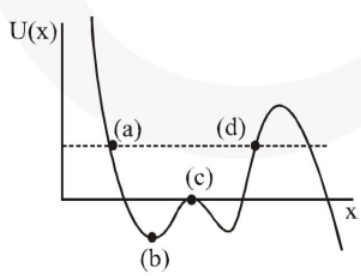


Column I	**Column II**
(A) By gravity	(P) 144 J
(B) By normal reaction	(Q) 32 J
(C) By friction	(R) 56 J
(D) By all the forces	(S) 48 J
	(T) None

Q17. In vertical circular motion of a bob, match the entries of column–I with entries of column–II. Here v_0 is the velocity of bob at lowest point & T is tension in string.

Column – I (Speed at lowest point)	**Column–II (Possible situation)**
(A) $v_0 = \sqrt{5g\ell}$	(P) $T_{lowest} - T_{highest} = 6mg$
(B) $v_0 = \sqrt{g\ell}$	(Q) String will slack for a finite time
(C) $v_0 = 2\sqrt{g\ell}$	(R) bob will oscillate
(D) $v_0 = 3\sqrt{g\ell}$	(S) bob will complete the circle

Q18. The figure below shows a graph of potential energy U(x) versus position x for a particle executing one dimensional motion along the x axis. The total mechanical energy of the system is indicated by the dashed line. The particle is somewhere between points a & d. Slope of tangent at a & d is same.



Column I	**Column II**
(A) Points where magnitude of the force is maximum	(P) (a)
(B) Points where the kinetic energy is maximum	(Q) (b)
(C) Points where the velocity is zero	(R) (c)
(D) Points where the force is zero	(S) (d)

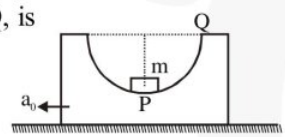
NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. A particle of mass 2kg moves on a smooth horizontal plane under the action of a single force $\vec{F} = (3\hat{i} + 4\hat{j})$ newton. Under this force it is displaced from (0,0) to (–1m, 1m). Find the work done by F (in Joule)

Q20. A body constrained to move along the z-axis of a coordinate system is subject to a constant force F given by $F = -\hat{i} + 2\hat{j} + 3\hat{k}$N where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the x-, y- and z-axis of the system respectively. What is the work done by this force in moving the body a distance of 4 m along the z-axis ?

Q21. A small block of mass m is lying at rest at point P of a wedge having a smooth semi circular track of radius R. The minimum value of horizontal acceleration a_0 (in m/s^2) of wedge so that mass can just reach the point Q, is



Q22. The potential function for a conservative force is given by U = (x + y). Find the work done by the conservative force in moving a particle from the point A(1, 1) to point B (2, 3).

Q23. The potential energy (in joules) function of a particle in a region of space is given as : $U = (2x^2 + 3y^3 + 2z)$ Here x, y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P (1m, 2m, 3m).

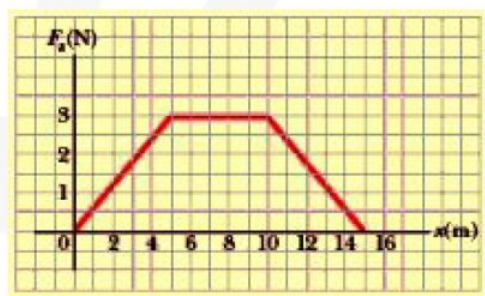
Q24. Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time t = 2 s and t = 4 s.

Q25. A block is released from rest from top of a rough curved track as shown in figure. It comes to rest at some point on the horizontal part. If its mass is 200 gm, calculate negative of work by friction in joules.



SUBJECTIVE TYPE

Q26. A particle is subject to a force F_x that varies with position as in figure. Find the work done by the force on the body as it moves
(a) from x = 0 to x = 5.00 m,
(b) from x = 5.00 m to x = 10.0 m, and
(c) from x = 10.0 m to x = 15.0 m.
(d) What is the total work done by the force over the distance x = 0 to x = 15.0 m?



Q27. The sign of work done by a force on a body is important to understand. State carefully if the following quantities are positive or negative:
(A) work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket.
(B) work done by gravitational force in the above case,
(C) work done by friction on a body sliding down an inclined plane,
(D) work done by an applied force on a body moving on a rough horizontal plane with uniform velocity,
(E) work done by the resistive force of air on a vibrating pendulum in bringing it to rest.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A block of mass m moves up on an inclined plane of angle θ. If μ is the co-efficient of friction between the block and the inclined plane, the work done in moving the block through a distance x along the plane is given by -

(A) mgx (sinθ + μcosθ)

(B) mgx sinθ

(C) mgx (sinθ – μcosθ)

(D) μmgx cosθ

Q2. **Statement 1:** The work done in pushing a block is more than the work done in pulling the block in a rough surface.

Statement 2: In the pushing condition more normal reaction increases the frictional force.

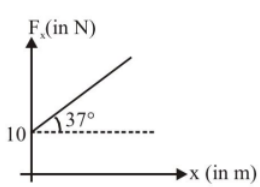
(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.

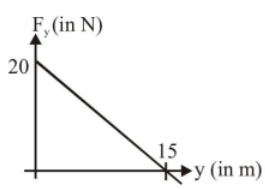
(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for Statement–I.

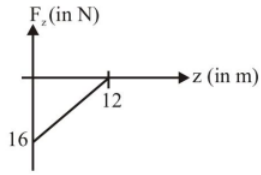
(C) Statement–I is true, Statement–II is false.

(D) Statement–I is false, Statement–II is true.

Q3. The components of a force acting on a particle are varying according to the graphs shown. To reach at point B (8, 20, 0) from point A(0, 5, 12) the particle moves on paths parallel to x-axis then y-axis and then z-axis, then work done by this force is







(A) 192 J (B) 58 J

(C) 250 J (D) 125 J

Q4. A force $\vec{F} = -k(y\hat{i} + x\hat{j})$, where k is a constant, acts on a particle moving in the x-y plane. Starting from the origin, the particle is taken from the origin (0, 0) to the point (a, 0) and then from the point (a, 0) to the point (a, a). The total work done by the force $\vec{F}$ on the particle is :

(A) ka^2 (B) $-ka^2$

(C) $2ka^2$ (D) $-2ka^2$

Q5. When a rubber-band is stretched by a distance x, it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:-

(A) $\frac{aL^2}{2} + \frac{bL^3}{3}$ (B) $\frac{1}{2}\left(\frac{aL^2}{2} + \frac{bL^3}{3}\right)$

(C) $aL^2 + bL^3$ (D) $\frac{1}{2}(aL^2 + bL^3)$

Q6. A spring of force constant 800 N/m has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is-

(A) 16 J (B) 8 J (C) 32 J (D) 24 J

Q7. This question has Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

If two springs S_1 and S_2 of force constants k_1 and k_2, respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2.

Statement-1: If stretched by the same amount, work done on S_1, will be more than that on S_2

Statement-2 : $k_1 < k_2$.

(A) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-1.

(B) Statement-1 is false, Statement-2 is true

(C) Statement-1 is true, Statement-2 is false

(D) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1.

Q8. A particle moves on a rough horizontal ground with some initial velocity say v_0. If $\frac{3}{4}$ of its kinetic energy is lost due to friction in time t_0 then coefficient of friction between the particle and the ground is :

(A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$ (C) $\frac{3v_0}{4gt_0}$ (D) $\frac{v_0}{gt_0}$

Q9. A small sphere starts falling from a very large height and after falling a distance of 100 m it attains the terminal velocity and continues to fall with this velocity. The work done by the atmosphere during the first fall of 100m is :

(A) Greater than the work done for next fall of 100 m

(B) Less than the work done for next fall of 100 m

(C) Equal to 100 mg

(D) Greater than 100 mg

Q10. A car of mass m starts moving so that its velocity varies according to the law $v = a\sqrt{s}$, where a is a constant, and s is the distance covered. The total work performed by all the forces which are acting on the car during the first t seconds after the beginning of motion is :

(A) $ma^4t^2/8$ (B) $ma^2t^4/8$

(C) $ma^4t^2/4$ (D) $ma^2t^4/4$

Q11. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to-

(A) x^2 (B) e^x

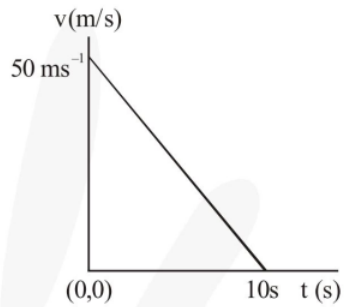
(C) x (D) $\log_e x$

Q12. At time t = 0s particle starts moving along the x-axis. If its kinetic energy increases uniformly with time 't', the net force acting on it must be proportional to :-

(A) $\sqrt{t}$ (B) constant

(C) t (D) $\frac{1}{\sqrt{t}}$

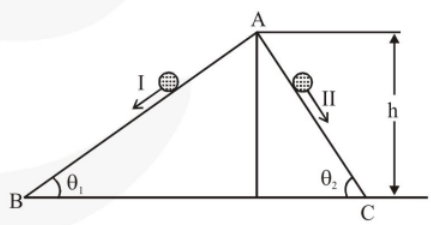
Q13. Velocity-time graph for a body of mass 10 kg is shown in figure. Work-done on the body in first two seconds of the motion is:-



(A) – 9300 J (B) – 12000 J

(C) 12000 J (D) – 4500 J

Q14. Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the following statement is correct?



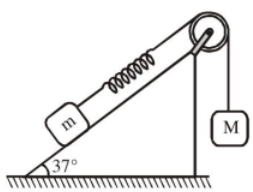
(A) Both the stones reach the bottom at the same time but not with the same speed.

(B) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.

(C) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.

(D) Both the stones reach the bottom at different times and with different speeds.

Q15. A block of mass m is attached with a massless spring of force constant k. The block is placed over a rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. System is released from rest when the spring was unstretched. The minimum value of M required to move the block up the plane is : (Neglect mass of string and pulley and friction in pulley)



(A) $\frac{3}{5}$ m (B) $\frac{4}{5}$ m

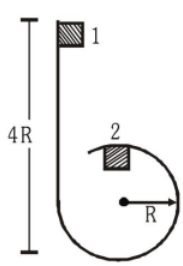
(C) 2 m (D) $\frac{3}{2}$ m

Q16. A simple pendulum has a string of length ℓ and bob of mass m. When the bob is at its lowest position, it is given the minimum horizontal speed necessary for it to move in a circular path about the point of suspension. The tension in the string at the lowest position of the bob is :

(A) 3mg (B) 4mg

(C) 5mg (D) 6mg

Q17. A cube of mass M starts at rest from point 1 at a height 4R, where R is the radius of the circular track. The cube slides down the frictionless track and around the loop. The force which the track exerts on the cube at point 2 is :



(A) 3 mg

(B) mg

(C) 2 mg

(D) cube will not reach the point 2.

Q18. A 2 kg block slides on a horizontal floor with a speed of 4 m/s. It strikes an uncompressed spring, and compresses it till the block is motionless. The kinetic friction force is 15 N and spring constant is 10, 000 N/m. The spring compresses by :-

(A) 5.5 cm (B) 2.5 cm

(C) 11.0 cm (D) 8.5 cm

Q19. **Statement 1 :** One end of ideal massless spring is connected to fixed vertical wall and other end to a block of mass m initially at rest on smooth horizontal surface. The spring is initially in natural length. Now a horizontal force F acts on block away from the wall. Then the maximum extension in spring is equal to maximum compression in spring.

Statement 2 : To compress or to expand an ideal unstretched spring by equal amount, same work is to be done on spring.

(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.

(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for statement–I

(C) Statement–I is true, Statement–II is false

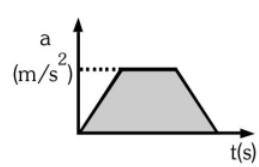
(D) Statement–I is false, Statement–II is true

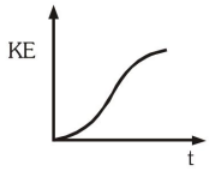
Q20. A mass of M kg is suspended by a weightless string. The horizontal force that is required to displace it until the string makes an angle of 45° with the initial vertical direction is-

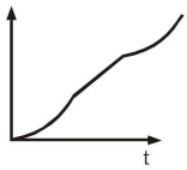
(A) $Mg(\sqrt{2}+1)$ (B) $Mg\sqrt{2}$

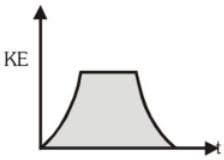
(C) $\frac{Mg}{\sqrt{2}}$ (D) $Mg(\sqrt{2}-1)$

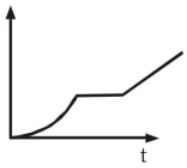
Q21. Acceleration versus time graph of a particle moving in a straight line is as shown in adjoining figure. If initially particle was at rest then corresponding kinetic energy versus time graph will be :



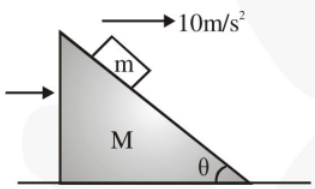
(A) 

(B) 

(C) 

(D) 

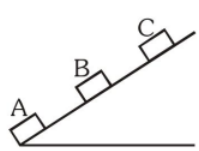
Q22. In the figure shown all the surfaces are frictionless, and mass of the block, m = 1 kg. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of 10 m/s^2 by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is :



(A) 30J (B) 60 J

(C) 150 J (D) 100 $\sqrt{3}$ J

Q23. A block has an initial kinetic energy of 128 J. It slides up from point A at the bottom of the inclined plane with uniform deceleration. When it passes point B, its kinetic energy is reduced by 80 J, and its mechanical energy is reduced by 35 J. Calculate the work done against friction when the block moves from A to the highest point C on the inclined plane.



(A) 42 J (B) 48 J

(C) 56 J (D) 72 J

Q24. A weight is hung freely from the end of a spring. A boy then slowly pushes the weight upwards until the spring becomes slack. The gain in gravitational potential energy of the weight during this process is equal to :

(A) The work done by the boy against the gravitational force acting on the weight.

(B) The loss of the stored energy by the spring minus the work done by the tension in the spring.

(C) The work done on the weight by the boy plus the stored energy lost by the spring.

(D) The work done on the weight by the boy minus the workdone by the tension in the spring plus the stored energy lost by the spring.

Q25. An object of mass m slides down a hill of height h of arbitrary shape and after travelling a certain horizontal path stops because of friction. The total vertical height decended is h. The friction coefficient is different for different segments for the entire path but is independent of the velocity and direction of motion. The work that a tangential force must perform to return the object to its initial position along the same path is :

(A) mgh (B) 2mgh

(C) 4mgh (D) –mgh

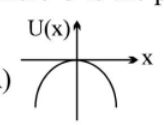
Q26. A particle is moved from (0, 0) to (a, a) under a force $\vec{F} = (3\hat{i} + 4\hat{j})$ from two paths. Path 1 is OP and path 2 is OQP. Let W_1 and W_2 be the work done by this force in these two paths. Then :

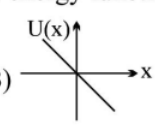
(A) $W_1 = W_2$ (B) $W_1 = 2W_2$

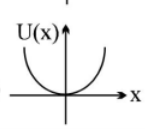
(C) $W_2 = 2W_1$ (D) $W_2 = 4W_1$

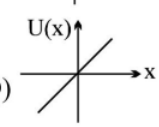
Q27. A particle is placed at the origin and a force F = kx is acting on it (where k is a positive constant). If U(0) = 0, the graph of U(x) versus x will be

(where U is the potential energy function)

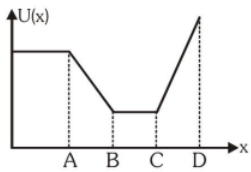
(A) 

(B) 

(C) 

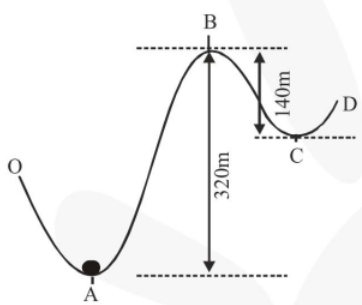
(D) 

Q28. As a particle moves along the x–axis it is acted upon by a conservative force. The potential energy is shown below as a function of the coordinate x of the particle. Rank the labelled regions according to the magnitude of the force, least to greatest.



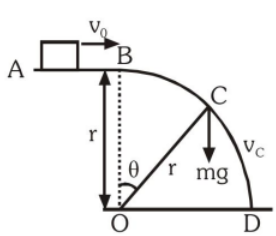
(A) AB, BC, CD (B) AB, CD, BC
(C) BC, CD, AB (D) BC, AB, CD

Q29. Track OABCD (as shown is figure) is smooth and fixed in vertical plane. What minimum speed has to be given to a particle lying at point A, so that it can reach point C?



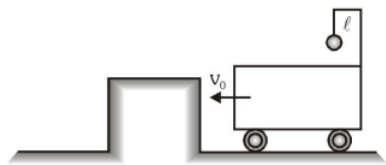
(A) 60 m/s (B) 100 m/s
(C) 70 m/s (D) 80 m/s

Q30. A small block slides with velocity $0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the figure. The block leaves the surface at point C. The angle θ in the figure is :



(A) $\cos^{-1}\frac{4}{9}$ (B) $\cos^{-1}\frac{3}{4}$
(C) $\cos^{-1}\frac{1}{2}$ (D) $\cos^{-1}\frac{4}{5}$

Q31. A bob is suspended from a crane by a cable of length $\ell = 5$ m. The crane and load are moving at a constant speed v_0. The crane is stopped by a bumper and the bob on the cable swings out an angle of 60°. The initial speed v_0 is– ($g = 9.8$ m/s^2)

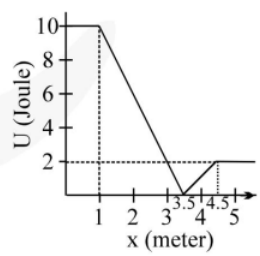


(A) 10 m/s (B) 7 m/s
(C) 4 m/s (D) 2 m/s

Q32. The potential energy of a 1 kg particle free to move along the x-axis is given by $U(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)$J. The total mechanical energy of the particle is 2 J. Then, the maximum speed (in m/s) is-

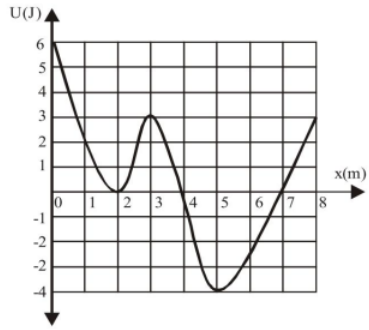
(A) $3/\sqrt{2}$ (B) $\sqrt{2}$
(C) $1/\sqrt{2}$ (D) 2

Q33. A body with mass 2 kg moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at x = 2m, then its speed when it crosses x = 5 m is



(A) zero (B) 1 ms^{-1}
(C) 2 ms^{-1} (D) 3 ms^{-1}

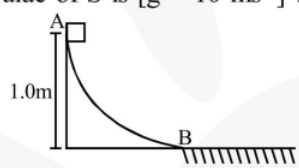
Q34. Potential energy curve U of a particle as function of the position of a particle is shown. The particle has total mechanical energy E of 3.0 joules.



(A) It can never be present at x= 0 m.
(B) It can never be present at x=5 m
(C) At x=2 its kinetic energy is 0 J
(D) At x =1 its kinetic energy 3 J

Q35. A block weighing 10 N travels down a smooth curved track AB joined to a rough horizontal surface (see figure). The rough surface has a friction coefficient of 0.20 with the block. If the block starts slipping on the track from a point 1.0 m above the horizontal surface, then it would move a distance S on the rough surface. The value of S is [g = 10 ms^{-2}] :-

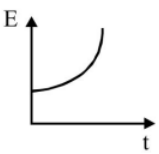
(A) 1m
(B) 2m
(C) 3m
(D) 5m

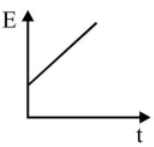


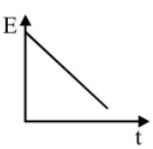
Q36. An ideal spring with spring constant K is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unstretched. The maximum extension in the spring is :

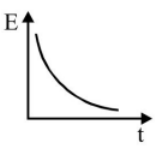
(A) $\frac{4Mg}{K}$ (B) $\frac{2Mg}{K}$ (C) $\frac{Mg}{K}$ (D) $\frac{Mg}{2K}$

Q37. A particle is dropped from a height h. A constant horizontal velocity is given to the particle. Taking g to be constant every where, kinetic energy E of the particle w.r.t. time t is correctly shown in :-

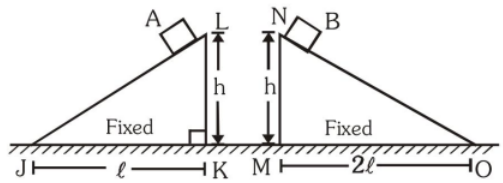
(A) 

(B) 

(C) 

(D) 

Q38. Two identical blocks A and B are placed on two inclined planes as shown in diagram. Neglect air resistance and other friction. Choose the correct statement :



Statement I: Kinetic energy of 'A' on sliding to J will be greater than the kinetic energy of B on falling to M.

Statement II: Acceleration of 'A' will be greater than acceleration of 'B' when both are released to slide on inclined plane.

Statement III : Work done by external agent to move block slowly from position B to O is negative

(A) statement I is true
(B) statement II is true
(C) statement I and III are true
(D) statement II and III are true

Q39. **Statement 1:** A particle is rotated in a vertical circle with the help of a string. Power produced by tension in the string on the particle is zero.

Statement 2 : Tension is always perpendicular to instantaneous velocity and Power = $\vec{F}.\vec{V}$.

(A) Statement–I is true, Statement–II is true; Statement–II is correct explanation for Statement–I.
(B) Statement–I is true, Statement–II is true; Statement–II is NOT a correct explanation for statement–I
(C) Statement–I is true, Statement–II is false
(D) Statement–I is false, Statement–II is true

Q40. A body is moved along a straight line by a machine delivering constant power. The distance moved by the body in time t is proportional to :

(A) $t^{1/2}$ (B) $t^{3/4}$
(C) $t^{3/2}$ (D) t^2

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements is TRUE for a system comprising of two bodies in contact exerting frictional force on each other:

(A) total work done by static friction on whole system is always zero.

(B) work done by static friction on a body is always zero

(C) work done by kinetic friction on a body is always negative

(D) total work done by internal kinetic friction on whole system is always negative

Q2. The kinetic energy of a particle continuously increase with time. Then

(A) the magnitude of its linear momentum is increasing continuously.

(B) its height above the ground must continuously decrease.

(C) the work done by all forces acting on the particle must be positive.

(D) the resultant force on the particle must be parallel to the velocity at all times.

Q3. A simple pendulum of length L and mass (bob) M is oscillating in a plane about a vertical line between angular limits $-\phi$ and ϕ. For an angular displacement θ, $[|\theta|<\phi]$ the tension in the string and velocity of the bob are T and v respectively. The following relations hold good under the above conditions.

(A) $T\cos\theta = Mg$

(B) $T - Mg\cos\theta = \frac{Mv^2}{L}$

(C) Tangential acceleration = $g\sin\theta$

(D) $T = Mg\cos\theta$

Q4. A particle is given a velocity at the bottom most position in a smooth spherical shell of radius 2m. It just complete a vertical circular motion. Then

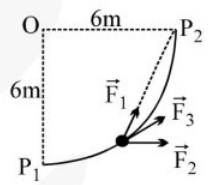
(A) Acceleration of particle when the velocity of the particle is in vertically upward direction is $g\sqrt{5}$

(B) Acceleration of particle when the velocity of the particle is in vertically downward direction is $g\sqrt{5}$

(C) Acceleration of the particle at the top most point of path is g

(D) Acceleration of the particle at the bottom most point is 5 g

Q5. A smooth track in the form of a quarter circle of radius 6 m lies in the vertical plane. A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. Force $\vec{F}_1$ is always toward P_2 and is always 20N in magnitude. Force $\vec{F}_2$ always acts horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. Select the correct alternative(s)



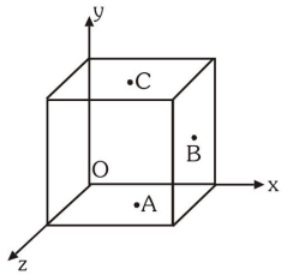
(A) work done by $\vec{F}_1$ is 120 J

(B) work done by $\vec{F}_2$ is 180 J

(C) work done by $\vec{F}_3$ is 45π

(D) $\vec{F}_1$ is conservative in nature

Q6. A particle is shifted from A to B and then from B to C where A, B and C are the mid-points of the corresponding faces of a cube of side 2m. If a force $\vec{F} = \left(3\hat{i} + 4\hat{j} - 5\hat{k}\right)$ N is continuously acting on the particle, then select correct alternative :-



(A) work done from A to B is 7 J

(B) work done from B to C is 1 J

(C) work done A to C is 8 J

(D) $\vec{F}$ is a conservative force

Q7. The potential energy of a particle moving along x–axis is given by U = 40 + 20 sin (4πx) where U is in J and x is in metre. Under the action of conservative force

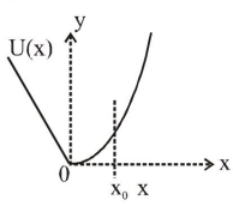
(A) If total mechanical energy is 60 J, then at $x = \frac{9}{8}$ m, particle is in equilibrium.

(B) If total mechanical energy is 60 J, then at $x = \frac{9}{8}$ m, particle is not in equilibrium.

(C) If total mechanical energy is 60 J, then at $x = \frac{5}{8}$m, particle's kinetic energy is minimum.

(D) If total mechanical energy is 60 J, then at $x = \frac{5}{8}$m, particle's kinetic energy is maximum.

Q8. A particle is released from rest at the position $x = x_0$. The potential energy graph of the particle is shown below.



$$U(x) = \begin{cases} -ax\ ;\ x < 0 \\ bx^2\ ;\ x > 0 \end{cases}$$

Which of the following statements is/are correct ?

(A) The subsequent motion is periodic.

(B) The subsequent motion is oscillatory

(C) The acceleration is a continuous function of time

(D) The subsequent motion is simple harmonic

Q9. A particle of mass m is at rest in a train moving with constant velocity with respect to ground. Now the particle is accelerated by a constant force F_0 acting along the direction of motion of train for time t_0. A girl in the train and a boy on the ground measure the work done by this force. Which of the following are **INCORRECT**?

(A) Both will measure the same work

(B) Boy will measure higher value than the girl

(C) Girl will measure higher value than the boy

(D) Data are insufficient for the measurement of work done by the force F_0

Q10. A particle of mass 2 kg is projected with an initial speed u = 10 m/sec at an angle θ = 30° with the horizontal

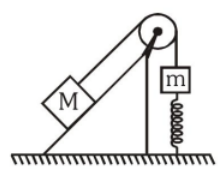
(A) Total work done on the particle during the first half of the total time of flight of the particle is (–25) J.

(B) Total work done on the particle during the total time of flight of the particle is 0J.

(C) Average power delivered to the particle during the first half of the flight is (–50) watt.

(D) The radius of curvature of the trajectory of the particle at the highest point of the projectile is 7.5m.

Q11. A large block of mass M on a smooth inclined plane is connected via a string to a small block m, which is attached to relaxed spring of force constant k with its other end attached to the ground (see diagram). Which of the following forces do negative work as the large block slides down the incline.



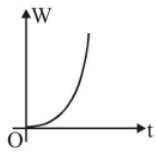
(A) The tension force on m.

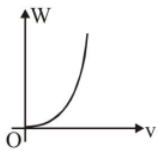
(B) The gravitational force on m.

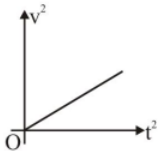
(C) The spring force on m.

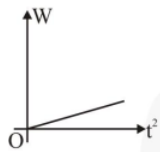
(D) The tension force on M.

Q12. A particle initially at rest on a smooth horizontal surface, is acted on by a constant horizontal force at time t =0. Then if W = work done, t = time and v = speed of the particle, the nearly correct graph(s) is/are :-

(A) 

(B) 

(C) 

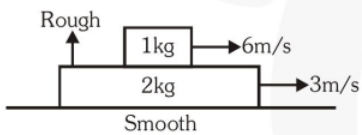
(D) 

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 13 & 14

In the figure shown upper block is given a velocity of 6 m/s and lower block 3 m/s. When relative motion between them is stopped.



Q13. (A) Work done by friction on upper block is negative

(B) Work done by friction on both the blocks is positive

(C) Work done by friction on lower block is negative

(D) Work done by friction on both the blocks is negative

Q14. (A) Work done by friction on upper block is – 10 J

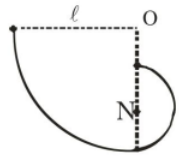
(B) Work done by friction on lower block is + 10 J

(C) Net work done by friction is zero

(D) All of the above

Paragraph for Question No. 15 to 17

The bob of simple pendulum of length ℓ is released from a point in the same horizontal line as the point of suspension (O) and at a distance ℓ from it.



Q15. The velocity of the bob at the lowest point of the string will be

(A) $v = \sqrt{2g\ell}$ (B) $v = \sqrt{g\ell}$

(C) $v = \sqrt{3g\ell}$ (D) $v = 2\sqrt{g\ell}$

Q16. If the string is catched by a nail located vertically below the point of suspension and the bob just swings around a complete circle around the nail, then the distance of the nail from point of suspension.

(A) $(2/5)\ell$ (B) $(2/3)\ell$

(C) $(3/5)\ell$ (D) $(1/3)\ell$

Q17. If the string of the pendulum is made of rubber then how much will it be stretched on reaching the bob at the lowest point.

(A) 2mg/k (B) 3mg/k

(C) 5 mg/k (D) mg/k

Paragraph for Question No. 18 & 19

A pendulum is suspended vertically from a point at rest. A force F in horizontal direction starts acting on it of constant magnitude $\frac{3}{4}Mg$ as shown. Pendulum bob is of mass m and string of length ℓ and acceleration due to gravity g.

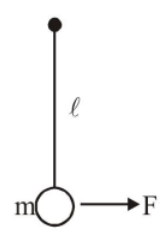
Q18. The angle made by string with vertical, when the speed of the bob is maximum, is :-

(A) 37°

(B) $\sin^{-1}\left(\frac{3}{5}\right)$

(C) $\cos^{-1}\left(\frac{4}{5}\right)$

(D) 53°



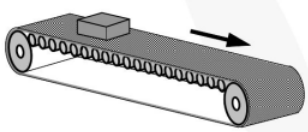
Q19. The minimum horizontal velocity that must be given to the bob at the initial position shown, such that the bob completes the circle is :-

(A) $\sqrt{5g\ell}$ (B) $\sqrt{\frac{25}{4}g\ell}$

(C) $\sqrt{\frac{24}{5}g\ell}$ (D) $\sqrt{\frac{23}{4}g\ell}$

Paragraph for Question 20 & 21

An object of mass M is gently placed on a horizontal conveyor belt, which is moving with uniform velocity v_o as shown in the figure. The coefficient of static friction is μ_s, the coefficient of kinetic friction is μ_k, and the acceleration of gravity is g. Initially the object slips for a while but finally moves without slipping together with the belt.



Q20. How far the conveyor belt moves while the object is slipping?

(A) $\frac{v_o^2}{\mu_k g}$ (B) $\frac{v_o^2}{2\mu_k g}$

(C) $\frac{v_o^2}{\mu_s g}$ (D) $\frac{v_o^2}{2\mu_s g}$

Q21. Work done on the object by friction force relative to the reference frame moving with the conveyer belt is

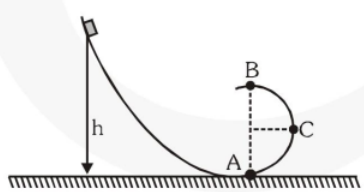
(A) $\frac{1}{2}Mv_o^2$ (B) $-\frac{1}{2}Mv_o^2$

(C) Mv_o^2 (D) zero

MATCH THE COLUMN TYPE

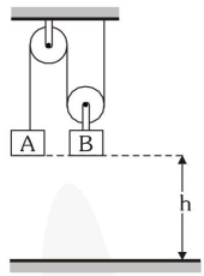
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q22. A block of mass m is released from top of a smooth track as shown in the figure. The end part of the track is a circle in vertical plane of radius R. N is normal reaction of the track at any point of the track. Match the entries of column I with entries of column–II.



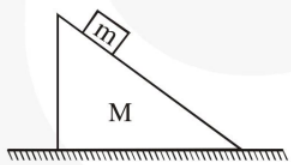
Column I	**Column II**
(A) $h = \frac{5}{2}R$	(P) Net force on the block at C is mg
(B) $h = \frac{9}{2}R$	(Q) $N_A - N_B = 6$ mg
(C) $h = R$	(R) Block leaves contact before B
(D) $h = 2R$	(S) Block will keep contact with the track is region between A & B.
	(T) $N_C > mg$

Q23. In the figure shown are two blocks A and B of same mass connected with pulley and string to each other. Initially both of them are at a height of h = 0.5 m from ground. After they are released they move in either direction and one of them strike the ground. For, the interval from releasing to when one of them strike, some physical quantities are in column I and their modulus values in SI units are in column II.



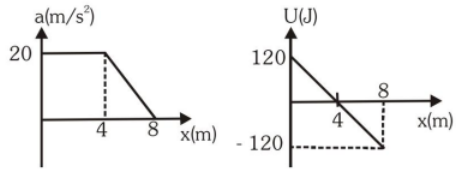
Column I	Column II
(A) Velocity of A immediately before any one of them strike ground.	(P) 1
(B) Velocity of B immediately before any one of them strike ground.	(Q) 2
(C) Ratio of work done by gravity on A to workdone by gravity on B.	(R) 3
(D) Acceleration of block A before any one of them strike ground.	(S) 4
	(T) 5

Q24. A block of mass m lies on wedge of mass M. The wedge in turn lies on smooth horizontal surface. Friction is absent everywhere. The wedge block system is released from rest. All situation given in column–I are to be estimated in duration the block undergoes a vertical displacement 'h' starting from rest (assume the block to be still on the wedge, g is acceleration due to gravity).



Column I	Column II
(A) Work done by normal reaction acting on the block is	(P) Positive
(B) Work done by normal reaction (exerted by block) acting on wedge is	(Q) Negative
(C) The sum of work done by normal reaction on block and work done by normal reaction (exerted by block) on wedge is	(R) Zero
(D) Net work done by all forces on block is	(S) Less than mgh in magnitude

Q25. Acceleration 'a' versus x and potential energy 'U' versus x graph of a particle moving along x–axis under some conservative and other external forces is as shown in figure. Mass of the particle is 1kg and velocity at x = 0 is 4 m/s. U-x graph is for conservative forces.At x = 8m :-

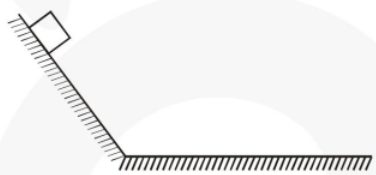


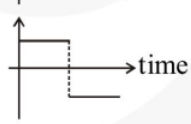
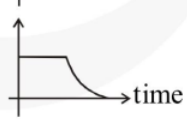
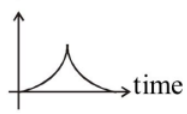
Column I	**Column II**
(A) Kinetic energy	(p) 120J
(B) Work done by conservative forces	(q) 240 J
(C) Total work done	(r) 128 J
(D) Work done by external forces	(s) 112 J
	(t) –120 J

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q26. A block slides down a smooth inclined plane & then moves on to a rough horizontal surface. List-I gives quantities related to block. List-II gives corresponding graphs of physical quantities versus time :-



List-I	**List-II**
(P) Velocity	(1) 
(Q) Acceleration	(2) 
(R) Mechanical energy	(3) 
(S) Kinetic energy	(4) 

Code :

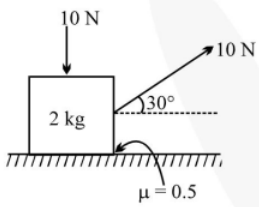
	P	**Q**	**R**	**S**
(A)	1	2	3	4
(B)	4	3	2	1
(C)	2	3	1	4
(D)	3	2	4	1

NUMERICAL TYPE

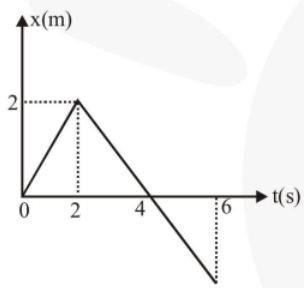
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q27. A point mass of 0.5 kg is moving along x-axis as $x = t^2 + 2t$, where, x is in meters and t is in seconds. Find the work done (in J) by all the forces acting on the body during the time interval [0, 2s].

Q28. A body of mass 2 kg is kept on a rough horizontal surface as shown in the figure. Find the work done by frictional force in the time interval t = 0 to t = 5 sec.



Q29. Position-time graph of a particle of mass 2 kg is shown in figure. Total work done on the particle from t = 0 to t = 4s is -

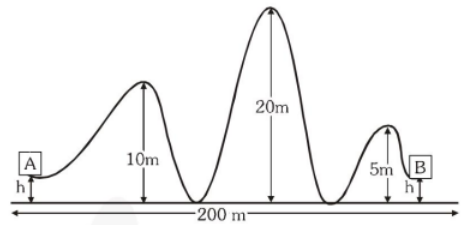


Q30. A force given by the relation F = 8 (N/s)t acts on a body of mass 2 kg initially at rest on a smooth surface where t is in seconds. Find the work done by this force on the body during first 2 seconds of its motion. Assume direction of force remain constant and body moves in the direction of force.

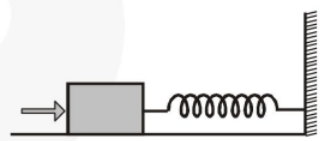
Q31. A spring, which is initially in its unstretched condition, is first stretched by a length x and then again by a further length x. The work done in the first case is W_1 and in the second case is W_2. Find $\frac{W_2}{W_1}$.

Q32. A point object of mass 2 kg is moved from point A to point B very slowly on a curved path by applying a tangential force on a curved path as shown in figure. Then find the work done by external force in moving the body. Given that $\mu_s = 0.3$, $\mu_k = 0.1$.

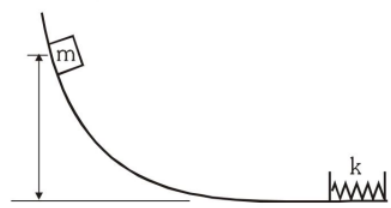
[g = 10 m/s²]



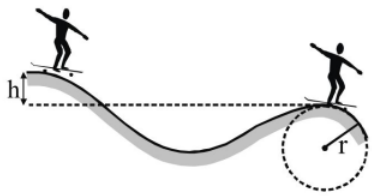
Q33. A block of mass 0.18 kg is attached to a spring of force-constant 2 N/m. The coefficient of friction between the block and the floor is 0.1. Initially the block is at rest and the spring is un-stretched. An impulse is given to the block as shown in the figure. The block slides a distance of 0.06 m and comes to rest for the first time. The initial velocity of the block in m/s is V= N/10. Then N is -



Q34. A mass m slides from rest at height h down a smooth curved surface which becomes horizontal at zero height (see figure). A spring is fixed horizontally on the level part of the surface. The spring constant is k N/m. When the mass encounters the spring it compresses it by an amount x = h/10. If m = 1 kg, h = 5m then find k /100.



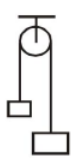
Q35. A skier starts from rest at the top of a hill. The skier coasts down the hill and up a second hill, as the drawing illustrates. The crest of the second hill is circular, with a radius of r = 36 m. Neglect friction and air resistance. What must be the height h (in m) of the first hill so that the skier just loses contact with the snow at the crest of the second hill?



Q36. An object of mass 4 kg falls from rest through a vertical distance of 10 m and reaches ground with a speed of 14 m/s. How much work is done against air friction in joules? [g = 10 m/s^2]

SUBJECTIVE TYPE

Q37. A light inextensible string that goes over a smooth fixed pulley as shown in the figure connects two blocks of masses 0.36 kg and 0.72 kg. Taking g = 10 m/s^2, find the work done (**in joules**) by the string on the block of mass 0.36 kg during the first second after the system is released from rest.



Q38. A body of mass 2 kg initially at rest moves under the action of an applied horizontal force of 7 N on a table with coefficient of kinetic friction = 0.1. Compute the

(A) work done by the applied force in 10 s,

(B) work done by friction in 10 s,

(C) work done by the net force on the body in 10 s,

(D) change in kinetic energy of the body in 10 s, and interpret your results.

Q39. A 4 kg particle moves along the X-axis. It's position x varies with time according to x(t) = t + 2t^3, where x is in m and t is in seconds. Compute:

(i) The kinetic energy at time t.

(ii) The force acting on the particle at time t.

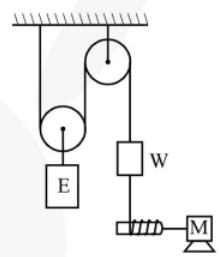
(iii) The power delivered to the particle at time t.

(iv) The work done on the particle from t = 0 to t = 2 seconds.

Q40. The elevator E has a mass of 3000 kg when fully loaded and is connected as shown to a counterweight W of mass 1000 kg. Determine the power in kilowatts delivered by the motor :

(a) when the elevator is moving down at a constant speed of 3 m/s,

(b) when it has an upward velocity of 3 m/s and a deceleration of 0.5 m/s^2.



Q41. A particle of mass m is hanging with the help of an elastic string of unstretched length a and force constant $\frac{mg}{a}$. The other end is fixed to a peg on vertical wall. String is given an additional extension of 2a in vertical downward direction by pulling the mass and released from rest. Find the maximum height reached by it during its subsequent motion above point of release.

(Neglect interaction with peg if any)

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	B	D	C	C	C	D	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	C	B	A	A	C	B	B	B	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**		
C	B	A	A	D	B	B	A		

Ex- 1A

1. B **2.** A,B,C **3.** A,C,D **4.** A,C **5.** A, D

6. A,D **7.** C **8.** D **9.** A **10.** B

11. D **12.** B **13.** B **14.** D

15. (A) PQR (B) PQR (C) R (D) RS

16. (A)–T, (B) –P, (C)–S, (D)–Q **17.** (A)–P,S (B) –R, (C)–Q, (D)–P,S

18. (A)-P, S, (B) -Q, (C)-P, S, (D)-Q, R **19.** 1 **20.** 12 J

21. 10 **22.** –3 **23.** 4 **24.** 46 J **25.** 4

26. (a) 7.5 J (b) 15 J (c) 7.5 J (d) 30 J

27. (A) +ve (B) –ve (C) –ve (D) +ve (E) –ve

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	C	B	A	B	B	A	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	D	D	**C**	**A**	**D**	**A**	A	D	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	C	C	**C**	**B**	A	A	D	D	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	A	C	A	D	B	A	D	**A**	C

Ex- 2A

1. A, D	2. A,C	3. B,C	4. C,D	5. B,C,D
6. A,B,C,D	7. A,C	8. A,B	9. A,C	10. A,B,C,D
11. B,C,D	12. A,B,C,D	13. A	14. A	15. A
16. C	17. B	18. A,B,C	19. D	20. A

21. B

22. (A) $\rightarrow$ (Q,S,T); (B) $\rightarrow$ (Q,S,T); (C) $\rightarrow$ (P); (D) $\rightarrow$ (R,T)

23. (A)$\rightarrow$(Q); (B)$\rightarrow$(P); (C)$\rightarrow$(Q); (D)$\rightarrow$(S)

24. (A) Q,S; (B) P,S; (C) R,S; (D) P,S

25. (A)–r, (B)–q, (C)–p, (D)–t

26. A	**27.** 8J	**28.** zero	**29.** 0	**30.** 64 J
31. 3	**32.** 400 J	**33.** 4	**34.** 4	**35.** 18m
36. 8	**37.** 8J			

38. (A) 875 J; (B) –250 J; (C) 625 J ; (D) 625 J

39. (i) $2 + 24t^2 + 72t^4$ J, (ii) 48 t N, (iii) $48t + 288t^3$ W, (iv) 1248 J

40. (a) –30 kW, 19.5 kW

41. $\frac{9a}{2}$